1) $y[n] = \frac{5}{6} y[n-1] - \frac{1}{6} y[n-2] + x[n]$

$x[n] = \delta[n] - \frac{1}{3}\delta[n-1]$

Taking Z transform:-

$Z\{y[n-k]\} = z^{-k} Y(z)$

$Y(z) = \frac{5}{6} z^{-1} Y(z) - \frac{1}{6} z^{-2} Y(z) + X(z)$

$Y(z)\left[1 - \frac{5}{6z} - \frac{1}{6z^2}\right] = X(z)$

$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - \frac{5}{6z} - \frac{1}{6z^2}} \Rightarrow \frac{z^2}{z^2 - \frac{5}{6}z - \frac{1}{6}}$

condition for stability ⇒ poles of $|| H(z) ||$ < 1

⇒ Poles of $H(z)$ → denominator = 0

$$z^2 - \frac{5}{6}z + \frac{1}{6} = 0$$

$$\left(z - \frac{1}{3}\right)\left(z - \frac{1}{2}\right) = 0$$

$$P_1 = 0.33 \qquad P_2 = 0.5$$

$$P_1 < 1 \qquad P_2 < 1$$

∴ System is stable.

$$y[n] = \frac{5}{6}y[n-1] - \frac{1}{6}y[n-2] + x(n)$$

$y[n]$ → depends on $y[n-1]$, $y[n-2]$ & $x[n]$

↳ so causal

$$x(n) = \delta[n] - \frac{1}{3}\delta[n-1].$$

$x[n]$ → depends on $\delta[n]$ & $\delta[n-1]$.

↳ so causal.